

The Body's Memory

How Unprocessed Emotional Trauma and Chronic Stress
Contribute to Inflammation and Autoimmune Disease

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Executive Summary

A growing body of scientific evidence demonstrates that unprocessed emotional trauma and chronic psychological stress are not merely psychological phenomena. They produce measurable and lasting changes in immune function, inflammatory signaling, and gene expression that significantly elevate the risk of autoimmune disease. This report traces the

biological pathways from traumatic experience through hypothalamic pituitary adrenal axis dysregulation, glucocorticoid receptor resistance, NF κB activation, pro-inflammatory cytokine cascades, epigenetic modification, and gut brain axis disruption to the onset and perpetuation of autoimmune pathology.

Drawing on research published through early 2026 in psychoneuroimmunology, neuroendocrinology, epigenetics, and microbiome science, this report synthesizes findings from landmark studies including the Kohler Forsberg et al. (2025) analysis of more than one hundred thousand women demonstrating a dose response relationship between adverse childhood experiences and autoimmune disease. It also incorporates the Chan et al. (2023) review of central regulation of stress evoked immune responses, the Webster et al. (2001) discovery of cytokine driven glucocorticoid receptor resistance, and the Banushi et al. (2025) review of intergenerational epigenetic transmission of trauma.

The findings converge on a central conclusion. The boundary between mental and physical health is a clinical convention rather than a biological reality. Trauma reshapes the immune system at the molecular, cellular, and systems level. The report also examines emerging evidence that mind-body therapeutic interventions including somatic experiencing, EMDR, yoga, vagal stimulation, and psychedelic assisted therapy may help reverse the physiological consequences of unresolved trauma and reduce autoimmune burden.

1. Introduction: The Body Remembers

"The body keeps the score." — Bessel van der Kolk, M.D.

The phrase “the body keeps the score,” popularized by psychiatrist Bessel van der Kolk in his landmark 2014 book, captures a central insight of modern psychoneuroimmunology. Traumatic experience is not stored only in memory but encoded in the body's physiological systems. For decades, clinicians observed that patients with histories of unresolved trauma and chronic stress presented with higher rates of autoimmune conditions, chronic pain, and inflammatory

disorders. What was once considered anecdotal is now supported by a robust and rapidly expanding body of research linking psychological trauma to measurable immune dysregulation.

This report examines the biological mechanisms through which unprocessed emotional trauma and chronic stress contribute to systemic inflammation and autoimmune disease. It draws on research in psychoneuroimmunology, the study of interactions among psychological processes, the nervous system, and the immune system, as well as neuroendocrinology, epigenetics, and the emerging science of the gut brain axis. Together, these disciplines are revealing that the pathways from traumatic experience to physical disease are specific, identifiable, and increasingly well characterized.

A note on terminology is essential. In this context, “unprocessed” trauma refers to a traumatic experience that has not been integrated or resolved. These experiences remain stored as fragmented sensory and somatic states that continue to activate the body's threat detection systems long after the original danger has passed. Unlike ordinary memories, which are consolidated into coherent narratives and gradually lose their emotional intensity, unprocessed traumatic memories retain their original vividness and physiological charge. They are not remembered so much as relived, triggered by environmental cues that reactivate the autonomic nervous system as if the threat were still present.

This sustained activation is the biological bridge between psychological experience and physical disease. When the body's stress response systems remain chronically engaged, when they never fully stand down, the resulting cascade of hormonal, neurological, and immunological changes creates the conditions for chronic inflammation and ultimately for the immune system to turn against the body's own tissues.

The following sections trace this cascade step by step, from the initial activation of the stress response to the molecular pathways that link cortisol dysregulation to autoimmune pathology, and from population level epidemiological evidence to the emerging science of epigenetic inheritance. The report concludes with an examination of therapeutic approaches that address

both the psychological roots and the physiological consequences of trauma related immune dysfunction.

2. The Stress Response System: The HPA Axis

The hypothalamic-pituitary-adrenal (HPA) axis is the body's central stress response system, an intricate neuroendocrine cascade, that translates the perception of threat into a coordinated physiological response. Understanding its normal function and its dysregulation under chronic stress is fundamental to grasping how psychological trauma produces physical disease.

Normal HPA Function

When the brain perceives a threat, whether physical, psychological, or social, the hypothalamus releases corticotropin-releasing hormone (CRH). CRH travels to the anterior pituitary gland, stimulating the release of adrenocorticotrophic hormone (ACTH) into the bloodstream. ACTH then signals the adrenal cortex (the outer layer of the adrenal glands, situated atop the kidneys) to produce and release cortisol, the body's primary stress hormone.

Under normal conditions, cortisol serves as a powerful anti-inflammatory agent and metabolic regulator. It suppresses unnecessary immune activity, mobilizes energy reserves, sharpens attention, and prepares the organism for action. Critically, cortisol also acts through a negative feedback loop: elevated cortisol levels signal the hypothalamus and pituitary to reduce CRH and ACTH production, thereby resolving the stress response once the threat has passed and returning the body to homeostasis.

The HPA Cascade

Hypothalamus (releases CRH) → **Anterior Pituitary** (releases ACTH) → **Adrenal Cortex** (produces Cortisol) → **Negative Feedback Loop** (cortisol signals hypothalamus and pituitary to reduce output)

Under chronic or traumatic stress, this feedback loop breaks down, leading to persistent cortisol dysregulation and impaired immune regulation.

Dysregulation Under Chronic Stress

When stress becomes chronic, or when traumatic experiences remain unresolved and continue to activate threat-detection circuitry, the HPA axis undergoes fundamental changes. The negative feedback loop that normally resolves the stress response begins to break down.

Research by Karin et al. (2020), published in *Molecular Systems Biology*, demonstrated that prolonged HPA activation leads to changes in the functional mass of hormone-secreting glands themselves. Their mathematical modeling of hormone dynamics revealed that the HPA axis operates on multiple timescales: while acute responses occur within minutes, chronic stress produces structural changes in gland tissue over weeks to months. These changes cause persistent cortisol dysregulation that outlasts the original stressor.

The result is paradoxical. Some individuals develop chronically elevated cortisol (hypercortisolism), while others develop blunted cortisol responses (hypocortisolism), a pattern frequently observed in individuals with post-traumatic stress disorder (PTSD) and histories of early-life trauma. Both patterns impair immune regulation, though through different mechanisms: hypercortisolism can suppress beneficial immune surveillance, while hypocortisolism removes the body's primary brake on inflammatory signaling.

A comprehensive 2025 review by Gutierrez Nunez et al., published in the *International Journal of Molecular Sciences*, synthesized the current evidence on chronic stress and autoimmunity. The authors documented that chronic stress leads to impaired HPA axis feedback,

glucocorticoid receptor resistance, and paradoxical cortisol dysregulation, fostering a pro-inflammatory state characterized by cytokine imbalance. This persistent inflammatory milieu, they concluded, shifts the immune response toward autoimmunity, the body's immune system attacking its own tissues.

3. From Acute Response to Chronic Inflammation

The acute stress response is adaptive and protective. It evolved to help organisms survive immediate threats, a function it performs with remarkable efficiency. The problem arises when a system designed for brief, intense activation becomes chronically engaged. The transition from acute stress response to chronic inflammation is the critical juncture at which psychological experience begins to produce physical disease.

The Protective-to-Destructive Shift

During an acute stress response, cortisol suppresses unnecessary immune activity, redirects metabolic energy toward muscles and brain, and temporarily enhances certain immune functions (such as the migration of immune cells to potential wound sites). This response is time-limited: once the threat passes, cortisol levels decline, the parasympathetic nervous system restores calm, and immune function returns to baseline.

When stress becomes chronic, whether from ongoing adversity, unresolved trauma, or persistent activation of threat-detection circuitry, this resolution never fully occurs. The immune system shifts from a state of readiness to a state of sustained low-grade activation, characterized by elevated levels of pro inflammatory cytokines: signaling molecules that drive and perpetuate inflammation.

Key Inflammatory Mediators

Three pro-inflammatory cytokines are central to this process:

- **Tumor Necrosis Factor Alpha (TNF- α):** A master regulator of inflammatory responses, capable of activating other immune cells and promoting tissue damage when chronically elevated.
- **Interleukin-6 (IL-6):** Involved in both pro-inflammatory and anti inflammatory signaling, but chronically elevated levels drive systemic inflammation, acute-phase protein production, and immune dysregulation.
- **Interleukin-1 β (IL-1 β):** A potent pro-inflammatory cytokine that promotes fever, immune cell activation, and inflammatory gene expression.

Peripheral Immune Remodeling

A landmark 2023 review by Chan, Poller, Swirski, and Russo, published in Nature Reviews Neuroscience, provided a comprehensive account of how the central nervous system regulates peripheral immune responses during stress. The authors reported that psychological stress induces alterations in hematopoiesis, the production of blood cells in the bone marrow, resulting in monocytosis (increased production of monocytes, a type of white blood cell), neutrophilia (elevated neutrophils), and lymphocytopenia (reduced lymphocytes). These shifts upregulate pro-inflammatory processes in immunologically relevant peripheral tissues.

Critically, Chan et al. documented that stress signals are conveyed from the CNS to immune cells via the autonomic nervous system and neuroendocrine pathways. Specific brain and spinal regions transmit signals to peripheral tissues, including bone marrow, spleen, gut, adipose tissue, and liver, to directly influence immunological function. This means that psychological states are not merely correlated with immune changes; they causally drive them through identifiable neural and hormonal pathways.

The authors further found that this peripheral inflammation contributes not only to physical comorbidities, metabolic syndrome, cardiovascular disease, immunosuppression, but also feeds back to the brain, contributing to psychiatric symptoms including depression and anxiety. This creates a bidirectional cycle in which psychological distress and immune dysregulation reinforce each other.

Trauma-Specific Immune Signatures

Sun, Qu, and Zhu (2021), in a review published in *Frontiers in Psychiatry* examining 65 studies, found that PTSD is associated with increased inflammatory factors and reduced anti-inflammatory factors. Their analysis revealed that traumatic stress disorders and immune disease share a common genetic basis at the gene expression level, suggesting that the relationship between trauma and immune dysfunction is not merely functional but rooted in shared molecular pathways.

4. Glucocorticoid Resistance and the NF- κ B Pathway

If HPA axis dysregulation is the starting point of trauma-related immune dysfunction, glucocorticoid receptor resistance is its critical amplifier. This section describes the molecular mechanism through which chronic stress creates a self-perpetuating inflammatory cycle, one that persists even when cortisol levels are adequate or elevated.

Normal Glucocorticoid Signaling

Under normal conditions, cortisol exerts its anti-inflammatory effects by binding to glucocorticoid receptors (GRs) inside immune cells. Once activated, the glucocorticoid receptor complex translocates to the cell nucleus, where it suppresses the transcription of

inflammatory genes. One of its primary targets is NF- κ B (nuclear factor kappa-light-chain-enhancer of activated B cells), a transcription factor that functions as a master regulator of inflammation.

When NF- κ B is activated, it drives the expression of pro-inflammatory cytokines (TNF- α , IL-6, IL-1 β), chemokines (which recruit immune cells to tissues), adhesion molecules (which allow immune cells to enter tissues), and other mediators of inflammation. In a healthy system, cortisol binding to GRs keeps NF- κ B in check, preventing excessive or prolonged inflammatory responses.

The Resistance Mechanism

A landmark study by Webster, Oakley, Jewell, and Cidlowski (2001), published in the Proceedings of the National Academy of Sciences, revealed a critical mechanism by which this regulatory system breaks down. The researchers demonstrated that pro-inflammatory cytokines, specifically TNF- α and IL-1, regulate human glucocorticoid receptor gene expression. Under inflammatory conditions, these cytokines disproportionately increase the production of the GR β isoform of the glucocorticoid receptor.

The GR β isoform is functionally significant because it does not bind cortisol. Moreover, GR β acts as a dominant negative inhibitor of the functional GR α isoform, meaning that it actively blocks the cortisol-responsive receptor from doing its job. The result is glucocorticoid resistance: the body continues to produce cortisol, but immune cells become progressively less responsive to its anti-inflammatory effects.

More recent work by Lockett, Inder, and Clifton (2024), published in Endocrine Reviews, has expanded our understanding of GR isoform diversity and its contribution to tissue-specific glucocorticoid sensitivity and resistance. Their review documented that alternative splicing, post-translational modifications, and tissue specific expression patterns create a complex landscape of glucocorticoid responsiveness, helping explain why chronic stress affects some tissues and organ systems more than others.

The Clinical Significance

Glucocorticoid resistance means that the body's primary endogenous mechanism for resolving inflammation is compromised. This is the molecular bridge between chronic psychological stress and sustained tissue-damaging inflammation. It explains a clinical paradox: patients with chronic stress or trauma histories often show both elevated cortisol and elevated inflammation, a pattern that would be contradictory if the cortisol-GR-NF- κ B pathway were functioning normally.

The Inflammatory Feedback Loop

Chronic Stress → Cortisol Dysregulation → **Glucocorticoid Receptor Resistance** ($\uparrow$ GR β , $\downarrow$ GR α function) → **Unchecked NF- κ B Activation** → Pro inflammatory Cytokines (TNF- α , IL-6, IL-1 β) → **Further GR Resistance** → Sustained Tissue Inflammation → **Autoimmune Risk**

This self-reinforcing cycle means that once glucocorticoid resistance is established, it can persist and worsen even without ongoing psychological stress, though continued stress accelerates the process.

5. Adverse Childhood Experiences and Autoimmune Risk

The biological mechanisms described in the preceding sections predict a specific epidemiological pattern: individuals exposed to significant psychological trauma, particularly during the critical developmental windows of childhood, should show elevated rates of autoimmune disease. The epidemiological evidence strongly confirms this prediction.

The Köhler-Forsberg et al. (2025) Study

A major 2025 study by Köhler-Forsberg et al., published in Psychological Medicine, provided some of the most comprehensive evidence to date. The researchers analyzed two large population-based cohorts: the Icelandic SAGA cohort (22,423 women) and the UK Biobank (86,492 women) , a combined sample of over 108,000 women.

Their key findings were striking:

- Adverse childhood experiences were remarkably prevalent: reported by 50% of participants in the SAGA cohort and 35% in the UK Biobank.
- ACEs were associated with increased autoimmune disease in a dose-response manner, with a prevalence ratio (PR) of 1.10 per additional ACE, meaning each additional type of adverse experience independently increased autoimmune risk.
- The strongest associations were observed for specific autoimmune conditions, as detailed in the table below.
- Sexual abuse and physical/emotional neglect were the ACE types most consistently associated with elevated autoimmune prevalence.
- Approximately one-quarter of the association between ACEs and autoimmune disease was statistically mediated through depression, anxiety, and PTSD, suggesting that both direct biological pathways and mental health sequelae contribute to autoimmune risk.

ACE-Associated Autoimmune Risk

Autoimmune Condition	Prevalence Ratio (PR)	Association Strength
Sjögren's Syndrome	1.34	Strongest
Polymyalgia Rheumatica	1.20	Strong
Rheumatoid Arthritis	1.14	Moderate

Systemic Lupus Erythematosus	1.13	Moderate
Thyroid Autoimmune Disease	1.11	Moderate

Source: Köhler-Forsberg et al. (2025), *Psychological Medicine*, 55, e36. Prevalence ratios represent increased risk per ACE.

The Thompson et al. (2026) Study

A 2026 study by Thompson et al., published in the *Journal of Child & Adolescent Trauma*, added an important dimension by examining the role of timing and severity. The researchers found that as ACE severity increased, autoimmune conditions increased significantly ($\beta = 0.47$, $p < .001$). Critically, early exposure (ages 0–5) had the strongest effect on later autoimmune disease compared to late exposure (ages 9–16), consistent with the understanding that early childhood is a period of heightened immunological and neuroendocrine plasticity.

The Dose-Response Relationship

The consistency of the dose-response relationship across studies, cohorts, and autoimmune conditions is particularly compelling. More types of adversity, more severe exposure, and earlier timing of exposure all correlate with greater autoimmune burden. This pattern is characteristic of a biological mechanism rather than a statistical artifact or confounding variable. It mirrors the dose-response logic of toxicology: the greater the exposure, the greater the biological effect.

6. Epigenetic Mechanisms: How Trauma Rewrites Gene Expression

Perhaps the most profound dimension of the trauma-inflammation connection is the discovery that traumatic experience can alter gene expression without changing the DNA sequence itself. These epigenetic modifications represent a biological mechanism through which life experience is written into the molecular architecture of cells, and potentially, transmitted across generations.

Three Primary Mechanisms

DNA Methylation. The most extensively studied epigenetic mechanism in trauma research is DNA methylation, the addition of methyl groups (CH_3) to cytosine bases in DNA, typically at CpG dinucleotide sites. Methylation of a gene's promoter region generally silences that gene, reducing or eliminating its expression. Trauma and chronic stress are associated with altered methylation patterns in several critical genes, most notably:

- **NR3C1:** The glucocorticoid receptor gene. Increased methylation of NR3C1 reduces the number of glucocorticoid receptors produced by immune and brain cells, directly contributing to the glucocorticoid resistance described in Section 4.
- **FKBP5:** A gene encoding a co-chaperone protein that regulates glucocorticoid receptor sensitivity. Demethylation of FKBP5 (increased expression) amplifies stress reactivity by making the GR less efficient, while also influencing immune cell function.

Histone Modifications. DNA is wrapped around histone proteins, forming a complex called chromatin. Chemical modifications to histones, including acetylation, methylation, phosphorylation, and ubiquitination, alter the accessibility of DNA to the transcriptional machinery. Stress-induced histone changes can make inflammatory genes more accessible (open chromatin) or stress regulatory genes less accessible (closed chromatin), shifting the

balance toward pro-inflammatory gene expression.

Non-Coding RNAs. A growing body of evidence implicates non-coding RNA molecules, particularly microRNAs (miRNAs), in the regulation of stress and immune gene expression. These small RNA molecules do not encode proteins but modulate gene expression post-transcriptionally by binding to messenger RNA and preventing its translation. Stress-altered miRNA profiles have been identified in individuals with PTSD and chronic stress, affecting pathways involved in inflammation, immune regulation, and neuroplasticity.

Intergenerational Transmission

Banushi, Collova, and Milroy (2025), in a review published in the *International Journal of Molecular Sciences*, examined the epigenetic mechanisms through which trauma effects are transmitted across generations. Their review highlighted DNA methylation of NR3C1 and FKBP5 as key molecular links between parental trauma exposure and altered stress-regulation biology in offspring. The authors documented evidence for both intergenerational transmission (from parent to child, potentially mediated by in-utero exposure and early caregiving) and the possibility of transgenerational transmission (effects persisting into the third generation or beyond, possibly through germline epigenetic changes).

A 2025 review in *OxJournal* further emphasized the central role of DNA methylation in stress response regulation and the significance of parental traumatic exposure, including prenatal stress, combat exposure, and childhood maltreatment, on offspring mental health and PTSD predisposition. These findings suggest that the immunological consequences of trauma may extend well beyond the individual who experienced it.

A Hopeful Finding

Critically, epigenetic modifications are not permanent in the way that genetic mutations are. They are potentially reversible. Environmental enrichment, psychotherapeutic interventions, pharmacological treatments, and mind-body practices have all shown the capacity to alter epigenetic marks in stress-related genes. This

reversibility represents one of the most important therapeutic implications of the trauma-inflammation model: the biological inscription of trauma, while durable, is not destiny.

7. The Gut-Brain Axis: Microbiome, Vagus Nerve, and Immunity

The gastrointestinal tract is far more than a digestive organ. It houses approximately 70% of the body's immune tissue, contains a complex enteric nervous system sometimes called the "second brain," and is home to trillions of microorganisms, bacteria, fungi, viruses, and archaea, collectively known as the microbiome. The bidirectional communication system linking the gut to the brain, known as the gut–brain axis, has emerged as a critical mediator of trauma-related immune dysregulation.

Bidirectional Communication Pathways

Han et al. (2022), in a review published in the *Journal of Inflammation Research*, documented the multiple pathways through which the gut and brain communicate:

- **The Vagus Nerve:** The primary neural conduit between the gut and the brain, carrying both afferent (gut-to-brain) and efferent (brain-to-gut) signals. The vagus nerve conveys information about gut microbial composition, intestinal inflammation, and nutrient status to the brainstem and higher brain regions.
- **The Neuroendocrine System:** Including the HPA axis, which both influences and is influenced by gut microbial activity. Gut bacteria produce and respond to stress hormones, creating a feedback loop between psychological stress and gastrointestinal function.
- **Cytokine Signaling:** Immune cells in the gut-associated lymphoid tissue (GALT)

produce cytokines that can enter the systemic circulation and influence brain function, while brain-derived inflammatory signals can alter gut immune activity.

- **Microbial Metabolites:** Gut bacteria produce a wide range of neuroactive compounds, including specific neurotransmitters (serotonin, GABA, dopamine), short-chain fatty acids (SCFAs), and tryptophan metabolites that influence both brain function and immune regulation.

How Trauma Disrupts the Gut-Brain Axis

Chronic stress and unresolved trauma disrupt the gut–brain axis through several interconnected mechanisms:

Reduced Vagal Tone. The vagus nerve mediates the body's anti-inflammatory reflex, a parasympathetic pathway through which the brain can suppress peripheral inflammation. Trauma and chronic stress reduce vagal tone (the nerve's regulatory capacity), weakening this anti-inflammatory brake and allowing peripheral inflammation to escalate.

Gut Dysbiosis. Chronic stress alters the composition of the gut microbiome, typically reducing microbial diversity and shifting the balance toward pro-inflammatory bacterial species. These changes affect the production of immunomodulatory metabolites, particularly short-chain fatty acids (butyrate, propionate, acetate) that normally support intestinal barrier integrity and anti-inflammatory immune responses.

Increased Intestinal Permeability. Stress-induced changes in gut barrier function, often colloquially termed "leaky gut", allow bacterial products (including lipopolysaccharide, or LPS) to translocate across the intestinal wall into the bloodstream. LPS is a potent activator of innate immune responses, triggering systemic inflammatory signaling through NF-κB and other pathways.

Connection to Autoimmunity

Loh et al. (2024), in a comprehensive review published in *Signal Transduction and Targeted Therapy* (Nature), documented the microbiota–gut–brain axis as an important regulator of glial functions, neural homeostasis, and immune balance. Their findings underscore that disruptions in this axis, whether initiated by psychological stress, dietary factors, or microbial imbalance, can contribute to the loss of immune tolerance: the body's ability to distinguish self from non-self. This loss of tolerance is the defining feature of autoimmune disease.

The gut–brain axis thus represents a critical convergence point for multiple trauma related pathways: HPA axis dysregulation, autonomic nervous system imbalance, inflammatory cytokine production, and microbial immune modulation all intersect in the gastrointestinal tract, creating a microenvironment uniquely vulnerable to stress-induced immune dysfunction.

8. Autoimmune Conditions Linked to Psychological Trauma

The preceding sections have described the biological mechanisms through which trauma and chronic stress can drive immune dysregulation. This section examines specific autoimmune conditions for which the trauma-inflammation connection has been established through clinical and epidemiological research.

Condition	Description	Trauma Link & Key Mechanism
Rheumatoid Arthritis (RA)	Chronic inflammatory disease targeting joint synovial tissue, causing pain, swelling, and progressive joint destruction.	Associated with ACEs (PR= 1.14). Linked to stress related HPA dysfunction and elevated TNF- α . Glucocorticoid resistance impairs the cortisol mediated suppression of joint inflammation.

Systemic Lupus Erythematosus (SLE)	Multi-system autoimmune disease causing the immune system to attack connective tissue, kidneys, skin, joints, and other organs.	Associated with ACEs (PR = 1.13). Animal and human studies link SLE to HPA axis dysregulation, NF-κB driven cytokine cascades, and epigenetic changes in immune regulatory genes.
Sjögren's Syndrome	Immune attack on moisture-producing glands (salivary, lacrimal), causing chronic dryness and systemic complications.	Showed the strongest ACE association (PR = 1.34). Suspected mechanisms include vagal dysfunction and stress-induced changes in glandular immune environments.
Thyroid Autoimmunity (Hashimoto's, Graves')	Immune-mediated destruction (Hashimoto's) or stimulation (Graves') of thyroid tissue, causing hypothyroidism or hyperthyroidism.	Associated with ACEs (PR = 1.11). Stress-related immune shifts and HPA thyroid axis interactions contribute to thyroid autoantibody production.
Multiple Sclerosis (MS)	Demyelinating disease in which the immune system attacks the myelin sheath surrounding nerve fibers in the CNS.	Associated with persistent HPA dysfunction in both human and animal studies. Stress-related increases in blood-brain barrier permeability may facilitate immune cell infiltration into the CNS.
Inflammatory Bowel Disease (IBD)	Chronic inflammatory conditions of the gastrointestinal tract, including Crohn's disease and ulcerative colitis.	Stress and trauma are associated with disease onset and flares. Gut–brain axis disruption, dysbiosis, and increased intestinal permeability are central mechanisms.

Psoriasis	Immune-mediated skin disease causing rapid skin cell proliferation, resulting in scaly, inflamed plaques.	Stress is a well documented trigger for flares. Linked to elevated TNF- α , IL-17, and IL-23 through NF- κ B-driven inflammatory cascades and neuroimmune skin signaling.
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Across these conditions, a common pattern emerges: trauma and chronic stress act through the convergent biological mechanisms described in this report, HPA dysregulation, glucocorticoid resistance, NF- κ B activation, cytokine imbalance, epigenetic modification, and gut–brain axis disruption, to create conditions favorable for the loss of immune tolerance and the development of autoimmune pathology. The specific organ or tissue affected likely reflects individual genetic predisposition, tissue-specific GR isoform expression, and the particular immunological vulnerabilities of each organ system.

9. Mind-Body Interventions: Pathways to Healing

If trauma is encoded in the body as well as the mind, effective treatment must address both dimensions. A growing body of evidence supports the efficacy of mind-body interventions that target the physiological consequences of trauma alongside its psychological manifestations.

Somatic Experiencing (SE)

Somatic Experiencing, developed by Peter Levine, is a body-oriented therapy that treats post-traumatic symptoms by working directly with interoceptive and proprioceptive sensations, the body's internal awareness of its own states. Kuhfuß et al. (2021), in a systematic review published in the European Journal of Psychotraumatology, evaluated the existing evidence for

SE and found positive effects on PTSD symptoms, affective and somatic symptoms, and overall well-being. The approach works by helping individuals gradually discharge the physiological energy that remains trapped in the nervous system following traumatic experiences, restoring the body's capacity for self-regulation.

EMDR (Eye Movement Desensitization and Reprocessing)

EMDR, extensively researched and advocated by Bessel van der Kolk and others, facilitates the processing and integration of traumatic memories through bilateral stimulation, typically guided eye movements performed while the individual attends to traumatic material. The therapy appears to help transform fragmented sensory and somatic trauma memories into coherent, integrated narratives, reducing their capacity to activate the body's threat-detection systems. EMDR is now recognized by the World Health Organization and multiple national clinical guidelines as a first-line treatment for PTSD.

Yoga and Mindfulness-Based Interventions

Body-centered practices including yoga therapy and mindfulness-based stress reduction (MBSR) have shown significant promise in addressing trauma-related symptoms. A survey conducted by PTSD UK in collaboration with the Mindful Institute found that 85% of participants with PTSD or complex PTSD reported symptom reduction when engaging in body-centered approaches. Yoga therapy addresses the interconnectedness of mind, body, and spirit through asana (physical postures), pranayama (breath regulation), and meditation, practices that directly influence vagal tone, autonomic nervous system balance, and cortisol regulation.

Vagal Tone Stimulation

Given the role of reduced vagal tone in trauma-related inflammation, practices that enhance

vagal function are of particular therapeutic interest. These include:

- Deep, slow diaphragmatic breathing (which directly stimulates vagal afferents)
- Cold water exposure (which activates the diving reflex and vagal output)
- Meditation and contemplative practices (which increase heart rate variability, a marker of vagal tone)
- Clinical vagus nerve stimulation (an emerging medical intervention using electrical or transcutaneous devices)

By strengthening the vagal anti-inflammatory reflex, these practices may help restore the body's capacity to regulate peripheral inflammation.

Psychedelic-Assisted Therapy

An emerging area of research involves the use of psychedelic-assisted therapy, including psilocybin, MDMA, and ketamine, for the treatment of trauma-related conditions. Banushi et al. (2025) noted that psychedelic-assisted treatments offer promising avenues to address both the psychological and potential epigenetic aspects of trauma. These substances appear to facilitate neuroplasticity and may enable the rapid reprocessing of traumatic material in therapeutic contexts, potentially reversing some of the neurobiological and immunological changes associated with chronic trauma.

The Integrative Principle

The evidence reviewed in this section points toward a central therapeutic principle: effective treatment of trauma-related inflammatory and autoimmune disease may require addressing both the psychological experience and the physiological cascade simultaneously. Interventions that work exclusively on the mind (talk therapy without somatic awareness) or exclusively on the body (immunosuppressive medication without trauma processing) may achieve partial results. Integrated approaches that recognize the unity of psychological and physiological trauma responses offer the most promising path toward comprehensive healing.

10. Conclusion: Toward an Integrated Understanding

The evidence reviewed in this report reveals that unprocessed emotional trauma and chronic stress are not merely risk factors for psychological distress, they are biological events that reshape the immune system at the molecular, cellular, and systems level. Through HPA axis dysregulation, glucocorticoid receptor resistance, NF- κ B-driven inflammatory cascades, epigenetic modifications, and gut-brain axis disruption, the body translates psychological suffering into physiological disease.

The pathways are now well characterized. Chronic stress disrupts the HPA axis, producing cortisol dysregulation that impairs immune regulation. Pro-inflammatory cytokines drive glucocorticoid receptor resistance through increased GR β expression, creating a self-reinforcing cycle of unchecked NF- κ B activation and sustained inflammation. Epigenetic modifications, particularly DNA methylation of the NR3C1 and FKBP5 genes, inscribe trauma's effects into gene expression patterns that may persist for years or even decades, and that can be transmitted to offspring. Disruption of the gut-brain axis compounds these effects through microbial dysbiosis, increased intestinal permeability, and reduced vagal anti inflammatory signaling.

The dose-response relationship between adverse childhood experiences and autoimmune risk, demonstrated across cohorts totaling over 100,000 individuals, provides compelling epidemiological confirmation of these mechanistic findings. The association is specific (certain ACE types and autoimmune conditions show stronger links), graded (more adversity produces greater risk), and temporally plausible (early-life exposure produces the strongest effects).

These findings carry profound clinical implications. The artificial boundary between "mental" and "physical" health, long enshrined in medical specialization, insurance coding, and clinical practice, is not supported by the biology. Autoimmune disease in patients with trauma histories cannot be fully understood or effectively treated without addressing the psychological experience that initiated the physiological cascade. Conversely, trauma therapy that ignores the

body's inflammatory and immunological state is incomplete.

The growing evidence for mind-body interventions, somatic experiencing, EMDR, yoga therapy, vagal stimulation, and psychedelic-assisted therapy, suggests that the biological inscription of trauma, while durable, is not permanent. Epigenetic modifications can be reversed. Vagal tone can be restored. Inflammatory cascades can be interrupted. The body's memory of trauma, while powerful, is not the final word.

Looking forward, the field is moving toward personalized, trauma-informed approaches to autoimmune care, treatment models that assess patients' trauma histories alongside their immunological profiles, that integrate psychological and medical interventions, and that recognize the individual as a unified biological system in which mind, brain, immune system, gut, and genome are continuously, intimately, and consequentially connected.

Healing the body may require healing the memory encoded within it.

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